

Securing Database Servers



With the ever-expanding data requirements for Web applications, database administrators often configure security parameters at the OS and database layer. Unfortunately, admins seldom consider implementing security at a network layer to protect the data from prying eyes. Recent cases of hacking into the IT infrastructure of finance firms show us that an increasing level of security awareness is required in this space, and this article aims to address this issue. It is targeted at systems and database administrators, as well as senior IT management staff, featuring case studies from a few popular open source databases.

For starters, there are seven layers that form the OSI networking model. Beginning at the physical layer, it goes up to the data-link, network, transport, session, presentation, and the application layers. While the physical-layer security is taken care of by surveillance systems, the application-layer security is handled by code instrumentation, which is the job of the developers. In the wake of overall security awareness in the IT world, responsible product managers incorporate code security as a mandated practice in their code-deployment release cycles.

All the intermediate five layers of the OSI model do require security measures too, by some means or the other, irrespective of the software component to which they are catering. Let's take a look at these layers, which are usually not taken too seriously by security administrators. It is important to note that these layers actually need to be more secure than the layers at the top and bottom, because there are such a variety of attacks on the interim layers that there is no single means, tool or process to control them.

The initial levels of compromising vulnerabilities usually start at the data-link and network layers

(alternatively called Layer-2 and Layer-3 attacks).

Layer-2 security is susceptible to MAC address spoofing, Layer-2 to flooding attacks, whereas Layer-3 security is subject to IP address-spoofing attacks. Similarly, Layers 4, 5 and 6 are prone to packet-crafting, session-stealing and cryptography-based attacks, respectively.

To understand this better, please refer to Figure 1, which shows various security methodologies and how each of these maps into the OSI layers of networking. While there is no such thing as 100 per cent security, the security design can incorporate carefully thought-out, customised security mechanisms, which can lead to a total security solution in an IT infrastructure. There is a wide range of devices from the security view-point, such as firewalls, intrusion-detection devices, content firewalls, etc, covering each layer, and also overlapping multiple network layers.

Why is typical security not enough for database servers?

Securing database servers needs administrators to go the extra mile in terms of technical design efforts. It is a job to be carried out by sys admins as well as database admins, to cover all the network layers. Often, the practice